# Formulae & Definitions — Topic 12 · Circular Motion

Radian 弧度 (rad) — the angle subtended at the centre of a circle by an arc of length equal to the radius. 360° = 2π rad, 1 rad ≈ 57.3°.

Angular displacement 角位移 θ — the angle swept out by an object with respect to the centre of rotation (rad).

Angular velocity 角速度 ω — the rate of change of angular displacement: $\omega = \Delta\theta/t = 2\pi/T$ (rad $s^{-1}$).

Linear velocity 线速度 v — the instantaneous (tangential) speed at a point on the path: $v = 2\pi r/T = r\theta/t = r\omega$. Its direction is tangential to the circle.

Uniform circular motion 匀速圆周运动 — constant speed but changing velocity (direction changes), so there is always an acceleration, a force and a change in momentum.

Centripetal acceleration 向心加速度 — $a = r\omega^2 = v^2/r$, directed towards the centre of the circle (m $s^{-2}$).

Centripetal force 向心力 — the resultant of all the real forces directed towards the centre: $F = mr\omega^2 = mv^2/r$ (N). It is at right angles to the velocity, so it changes direction but not speed.

点评 · Insight

重点：向心力不是一个具体性质的力（如重力、摩擦力），而是各具体力在指向圆心方向的合效果。解题关键是先分析向心力的来源——由哪些指向圆心的力（或分量）提供。